



Mostafa Sobhy

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EXPERIENCE

Senior Data Scientist – DHL, Bonn

FEB 2025 – PRESENT

- Building scalable ML solutions for global logistics optimization and supply chain analytics

Senior Data Scientist – Bain & Company, Düsseldorf

NOV 2021 – JAN 2025

- Led end-to-end development of GenAI solutions on Azure for banking clients, automating customer service workflows and financial analysis pipelines
- Architected production-ready RAG systems using LangChain and vector databases, reducing document processing time by 40%
- Built multi-modal mixture model for customer segmentation, achieving 6% lift over scikit-learn GMM baseline on mixed survey data
- Delivered unsupervised learning solutions across retail, energy, and finance verticals

Data Scientist – E.ON SE, Essen

JAN 2020 – OCT 2021

- Developed VAE-based predictive maintenance model for high-voltage transformers, increasing failure lead time by 20% and saving millions in downtime costs
- Designed multi-objective optimization framework using Model Predictive Control (MPC) for EV charging infrastructure integration

Research Assistant – Fraunhofer ISE, Freiburg

MAY 2017 – MAR 2019

- Built probabilistic simulation models for distributed energy systems under uncertainty

EDUCATION

M.Sc. Energy Engineering – University of Freiburg, Germany

2016 – 2019

B.Sc. Mechanical Engineering – Benha University, Egypt

2011 – 2016

CERTIFICATIONS

MicroMasters: Statistics & Data Science – MITx,

2021 – 2024

Machine Learning Specialization – Stanford Online,

2019 – NOW

CORE SKILLS

Machine Learning

Stack: PyTorch, scikit-learn

Modeling: representation learning (VAE), probabilistic models (GMM), clustering

NLP: Hugging Face Transformers

Time series: NeuralProphet

GenAI & LLMs

Fine-tuning & alignment: LoRA/PEFT, instruction tuning, RLHF, reward modeling

RAG: embeddings, vector stores, dense/sparse retrieval

Agents: LangChain, tool-use pipelines, planner/executor patterns

Data & MLOps

Data: SQL, PySpark, Polars

Orchestration & pipelines: Airflow, Kube-flow

Experiment tracking: MLflow

Cloud ML platforms: Azure ML, AWS SageMaker

Focus Areas

Generative AI & LLM systems (RAG, agents)
Unsupervised & probabilistic learning

LANGUAGES

English – Fluent (C2)

German – Professional (B2)

Arabic – Native